

DATA ANALYSIS & AI FUNDAMENTALS

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```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

# Load and explore data
df = pd.read_csv('data.csv')
print(df.head())
print(df.describe())

# Data preprocessing
df = df.dropna()
X = df.drop('target', axis=1)
y = df['target']

# Train test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

# Train model
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)

# Evaluate
score = model.score(X_test, y_test)
print(f"Model Accuracy: {score:.2f}")
```

DATA VISUALIZATION AND INSIGHTS EXTRACTION

- Univariate Analysis(Categorical or Continuous)
- Bivariate Analysis (Categorical or Continuous)
- Multivariate Analysis (Continuous)
- Advanced Plots and Charts

1. LOAD THE REQUIRED LIBRARIES

In [1]:

```
#!pip install mpl_toolkits
```

In [2]:

```
import pandas as pd                                ## For data importation and dataframes manipu
import numpy as np                                  ## For Numeric Manipulation
import seaborn as sns                               ## For data visualization
import matplotlib.pyplot as plt                     ## For data visualization
from mpl_toolkits.mpl3d import Axes3D              ## For 3D Visualization

### Import warnings
import warnings                                     ### For ignoring the warning messages
warnings.filterwarnings('ignore')
```

1.1. SET UP THE VISUALIZATION STYLE

In [3]:

```
sns.set_style("whitegrid")
plt.rcParams["figure.figsize"] = [10, 6]
plt.rcParams["font.size"] = 10
```

1.2. DATA IMPORTATION

In [4]:

```
df = pd.read_csv("GSSsubset.csv")
df.head()
```

Out[4]:

	id	sex	degree	income	marital	age	height	weight	hrswrk
0	1	MALE	BACHELOR	60967.50	DIVORCED	53	72	190	60
1	2	FEMALE	BACHELOR	60967.50	MARRIED	26	60	97	40
2	4	FEMALE	BACHELOR	10161.25	MARRIED	56	68	160	20
3	14	FEMALE	HIGH SCHOOL	17551.25	MARRIED	40	65	156	37
4	16	MALE	HIGH SCHOOL	17551.25	MARRIED	56	66	210	6

1.3. CHECK THE BASIC DATA INFORMATION

In [5]:

```
### Dimension of the Data
print("="*110)
print("CHECKING THE BASIC DATA INFORMATION")
print("="*110)
print(f"Display the dimension of the data: {df.shape}")
print(f"Display the column list: {list(df.columns)}")
print(f"Display the types of the Variables: {df.info()}")
```

CHECKING THE BASIC DATA INFORMATION

Display the dimension of the data: (994, 9)

Display the column list: ['id', 'sex', 'degree', 'income', 'marital', 'age', 'height', 'weight', 'hrswrk']

<class 'pandas.DataFrame'>

RangeIndex: 994 entries, 0 to 993

Data columns (total 9 columns):

#	Column	Non-Null Count	Dtype
0	id	994 non-null	int64
1	sex	994 non-null	str
2	degree	994 non-null	str
3	income	994 non-null	float64
4	marital	994 non-null	str
5	age	994 non-null	int64
6	height	994 non-null	int64
7	weight	994 non-null	int64
8	hrswrk	994 non-null	int64

dtypes: float64(1), int64(5), str(3)

memory usage: 93.7 KB

Display the types of the Variables: None

2. UNIVARIATE DATA VISUALIZATION

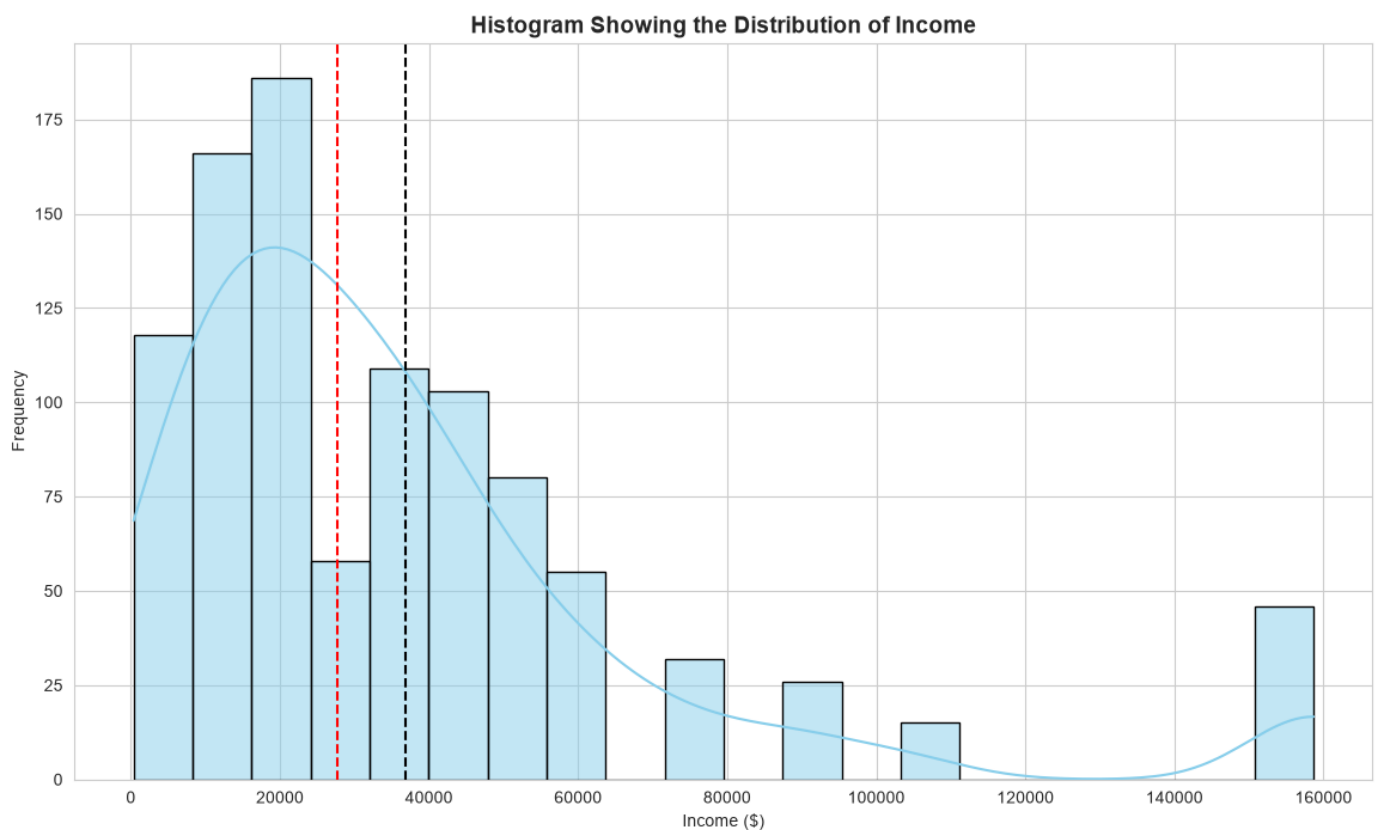
2.1. Histogram For the Distribution of Income

In [6]:

```
plt.figure(figsize = (14, 8))
sns.histplot(df['income'], bins = 20, kde = True,
             color = "skyblue", edgecolor = "black")
plt.title("Histogram Showing the Distribution of Income", fontsize = 14, fontweight = "b")
plt.xlabel("Income ($)")
plt.ylabel("Frequency")

plt.axvline(df['income'].median(), color = 'red',
            linestyle = '--',
            label = f'Median: ${df['income'].median():,.0f}')

plt.axvline(df['income'].mean(), color = 'black',
            linestyle = '--',
            label = f'Mean: ${df['income'].mean():,.0f}')
plt.show()
```



Comparing the mean and median is a quick way to identify skewness (asymmetry) in a dataset. Because the mean is sensitive to extreme outliers, it shifts toward the "tail" of the data, while the median represents the middle point.

1. Zero Skew (Symmetrical)

Rule: Mean = Median.

Shape: The distribution looks like a classic, symmetrical bell curve when graphed on a Scribbr Skewness Guide.

2. Right/Positive Skew

Rule: Mean > Median

Shape: The data features a long tail extending toward the higher, positive numbers. The extreme high values pull the mean up, while the median stays closer to the bulk of the data.

3. Left/Negative Skew

Rule: Mean < Median Shape: The data features a long tail extending toward the lower, negative numbers. The extreme low values pull the mean down, leaving the median higher than the mean.

In [7]:

```
df.drop(columns = ['id']).describe()
```

Out[7]:

	income	age	height	weight	hrswrk
count	994.000000	994.000000	994.000000	994.000000	994.000000
mean	36887.218352	44.488934	67.414487	181.318913	42.640845
std	34702.256068	13.126865	4.037741	41.641615	14.394974
min	369.500000	19.000000	57.000000	90.000000	1.000000
25%	15703.750000	33.000000	64.000000	150.000000	38.000000
50%	27712.500000	44.000000	67.000000	175.000000	40.000000
75%	49882.500000	55.000000	70.000000	205.000000	50.000000
max	158656.952000	79.000000	79.000000	410.000000	89.000000

2.2. Distribution of Income Using Box Plot

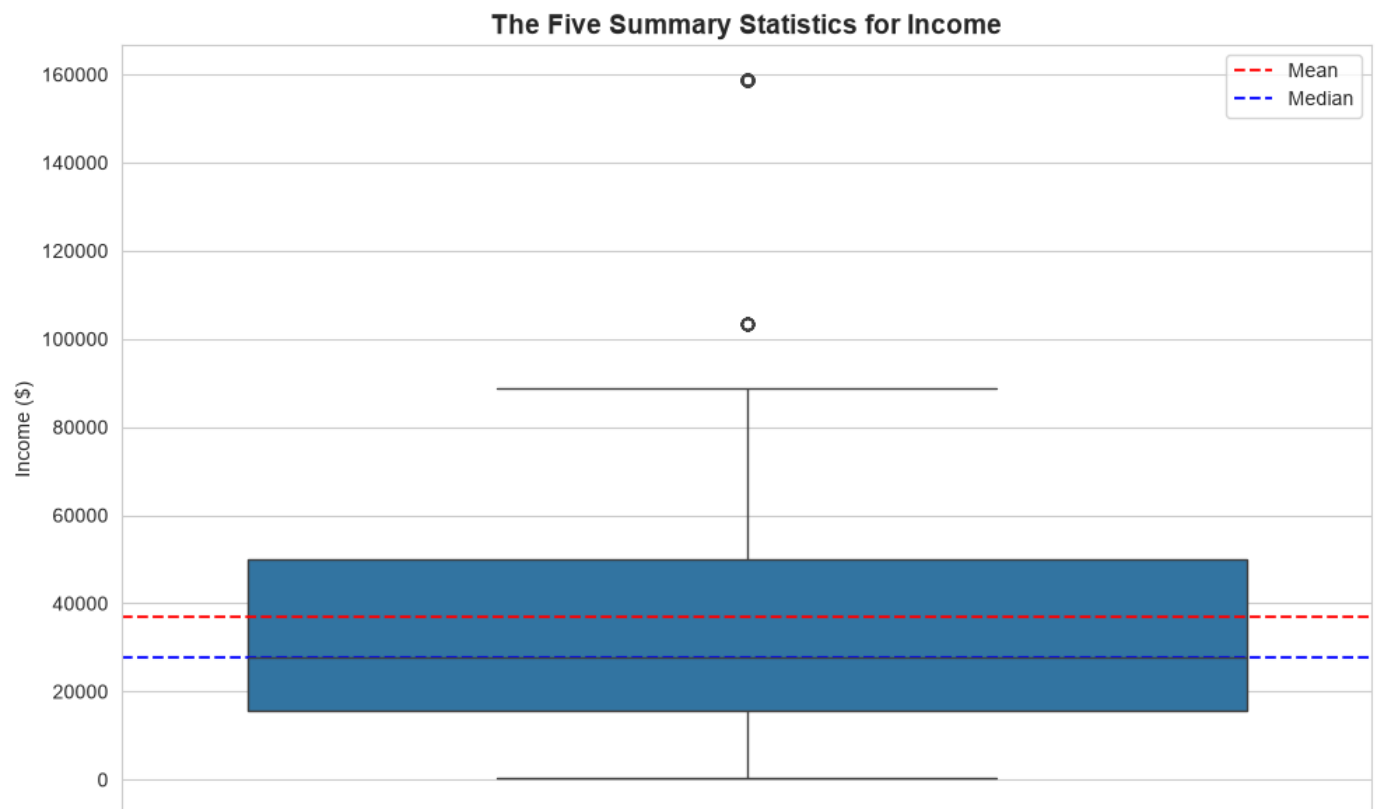
In [8]:

```
plt.figure(figsize = (10, 6))    ### Initialize the figure size
sns.boxplot(y = df['income'])    ### Pick the type of chart
plt.title("The Five Summary Statistics for Income",
          fontsize = 14,
          fontweight = 'bold')
plt.ylabel("Income ($)")

plt.axhline(df['income'].mean(), color = "red",
            linestyle = '--', alpha = 0.9, label = 'Mean')

plt.axhline(df['income'].median(), color = "blue",
            linestyle = '--', alpha = 0.9, label = 'Median')
plt.legend()
plt.tight_layout()

plt.show()                      ### Display the chart
```



2.3. Violin and Histogram for the Distribution of Ages

In [9]:

```
df.head()
```

Out[9]:

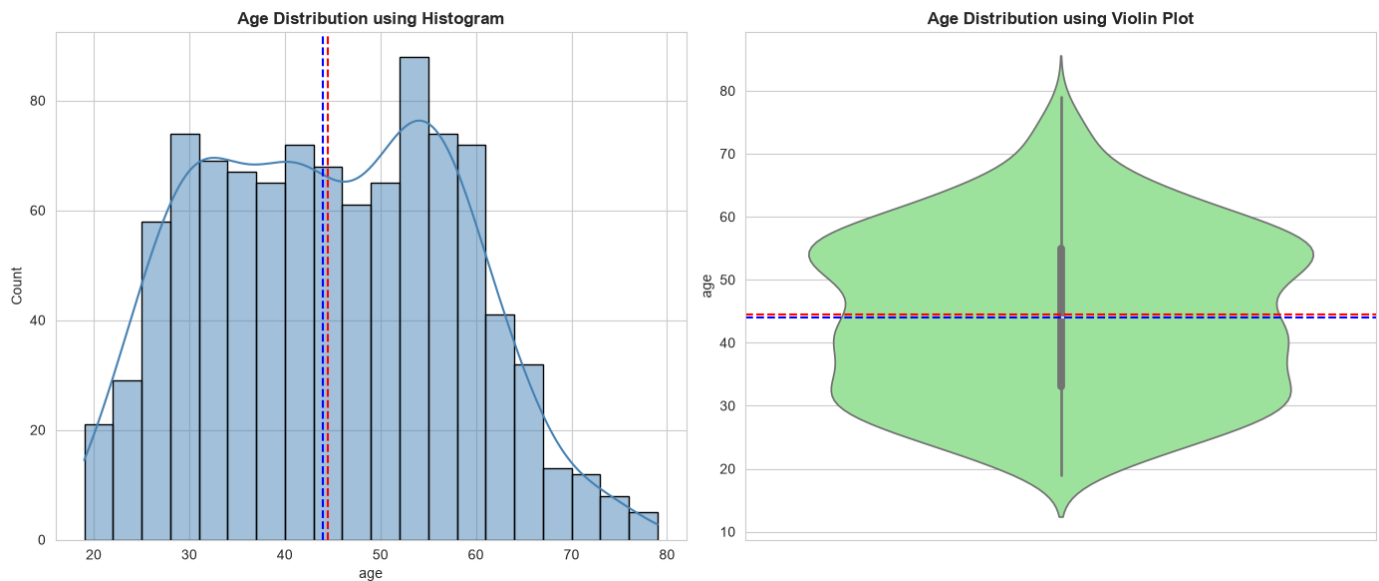
	id	sex	degree	income	marital	age	height	weight	hrswrk
0	1	MALE	BACHELOR	60967.50	DIVORCED	53	72	190	60
1	2	FEMALE	BACHELOR	60967.50	MARRIED	26	60	97	40
2	4	FEMALE	BACHELOR	10161.25	MARRIED	56	68	160	20
3	14	FEMALE	HIGH SCHOOL	17551.25	MARRIED	40	65	156	37
4	16	MALE	HIGH SCHOOL	17551.25	MARRIED	56	66	210	6

In [10]:

```
fig, axes = plt.subplots(1, 2, figsize = (14, 6))

sns.histplot(df['age'], bins = 20, kde = True, ax = axes[0],
             color = 'steelblue', edgecolor = 'black')
axes[0].set_title("Age Distribution using Histogram",
                  fontsize = 12,
                  fontweight = 'bold')
axes[0].axvline(df['age'].mean(), color = 'red',
                linestyle = '--',
                label = f'Mean: ${df['age'].mean():.0f}')
axes[0].axvline(df['age'].median(), color = 'blue',
                linestyle = '--',
                label = f'Median: ${df['age'].median():.0f}')
```

```
sns.violinplot(y = df['age'], ax = axes[1], color = 'lightgreen')
axes[1].set_title("Age Distribution using Violin Plot",
                  fontsize = 12,
                  fontweight = 'bold')
axes[1].axhline(df['age'].mean(), color = 'red',
                linestyle = '--',
                label = f'Mean: ${df['age'].mean():.0f}')
axes[1].axhline(df['age'].median(), color = 'blue',
                linestyle = '--',
                label = f'Median: ${df['age'].median():.0f}')
plt.tight_layout()
plt.show()
```

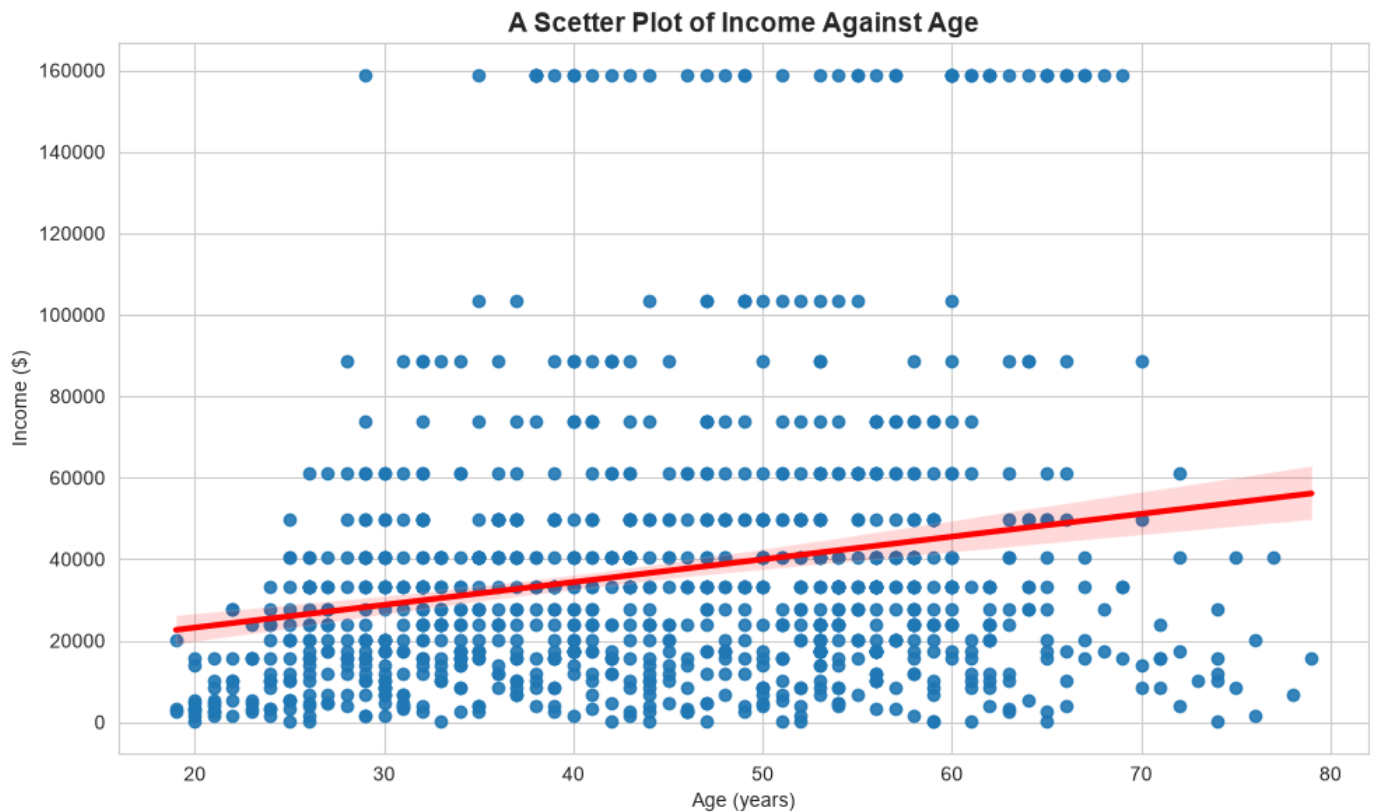


2.4. Scatter Plot Showing the Relationship Between Age and Income

In [11]:

```
plt.figure(figsize = (10, 6))
sns.regplot(data = df, x = 'age', y = 'income', scatter_kws = {'alpha': 0.9},
            line_kws = {'color': 'red', 'linewidth': 3})
plt.title("A Scetter Plot of Income Against Age",
          fontsize = 14,
          fontweight = "bold")
plt.xlabel("Age (years)")
plt.ylabel("Income ($)")

plt.tight_layout()
plt.show()
```

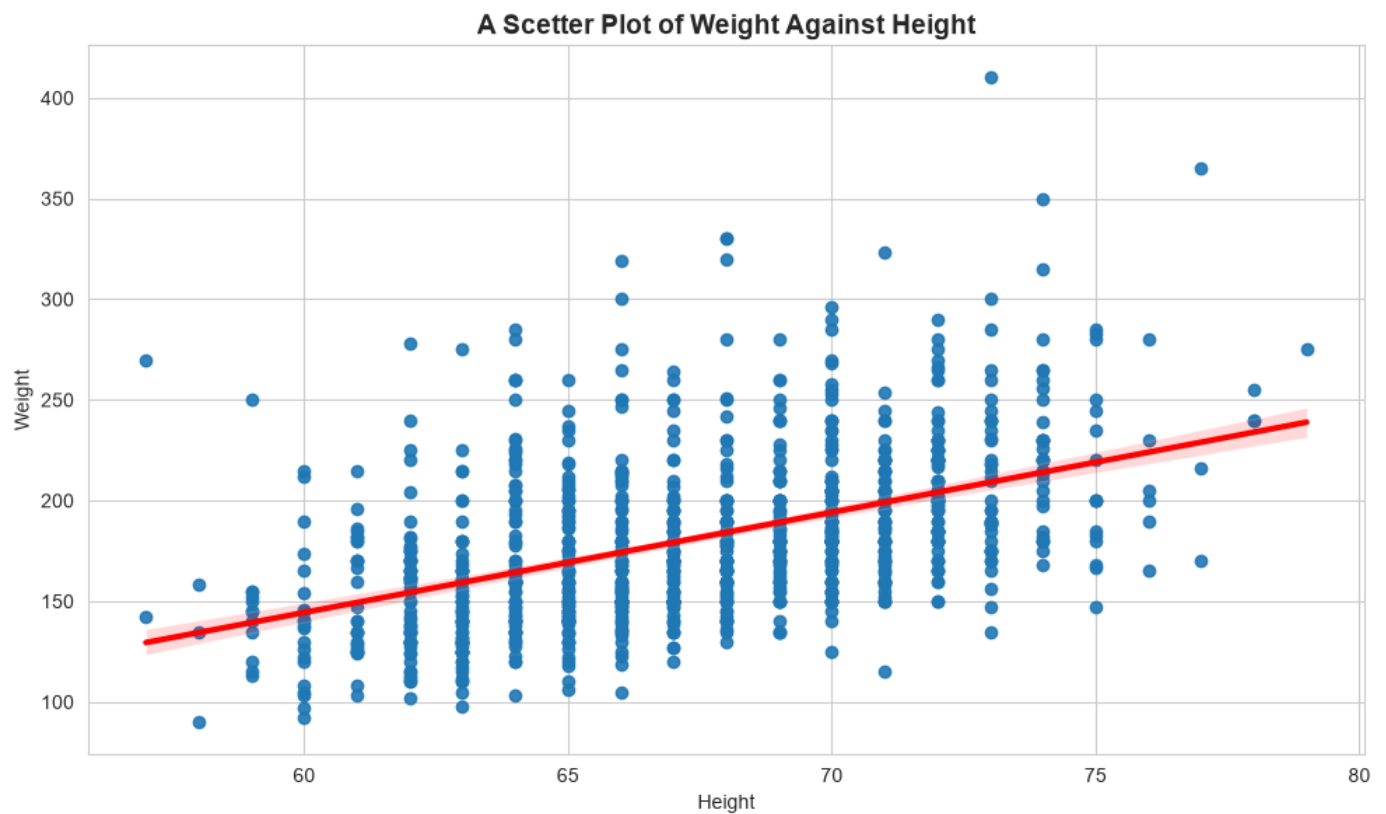


2.4. Scatter Plot Showing the Relationship Between Age and Income

In [12]:

```
plt.figure(figsize = (10, 6))
sns.regplot(data = df, x = 'height', y = 'weight', scatter_kws = {'alpha': 0.9},
            line_kws = {'color': 'red', 'linewidth': 3})
plt.title("A Scetter Plot of Weight Against Height",
          fontsize = 14,
          fontweight = "bold")
plt.xlabel("Height")
plt.ylabel("Weight")

plt.tight_layout()
plt.show()
```

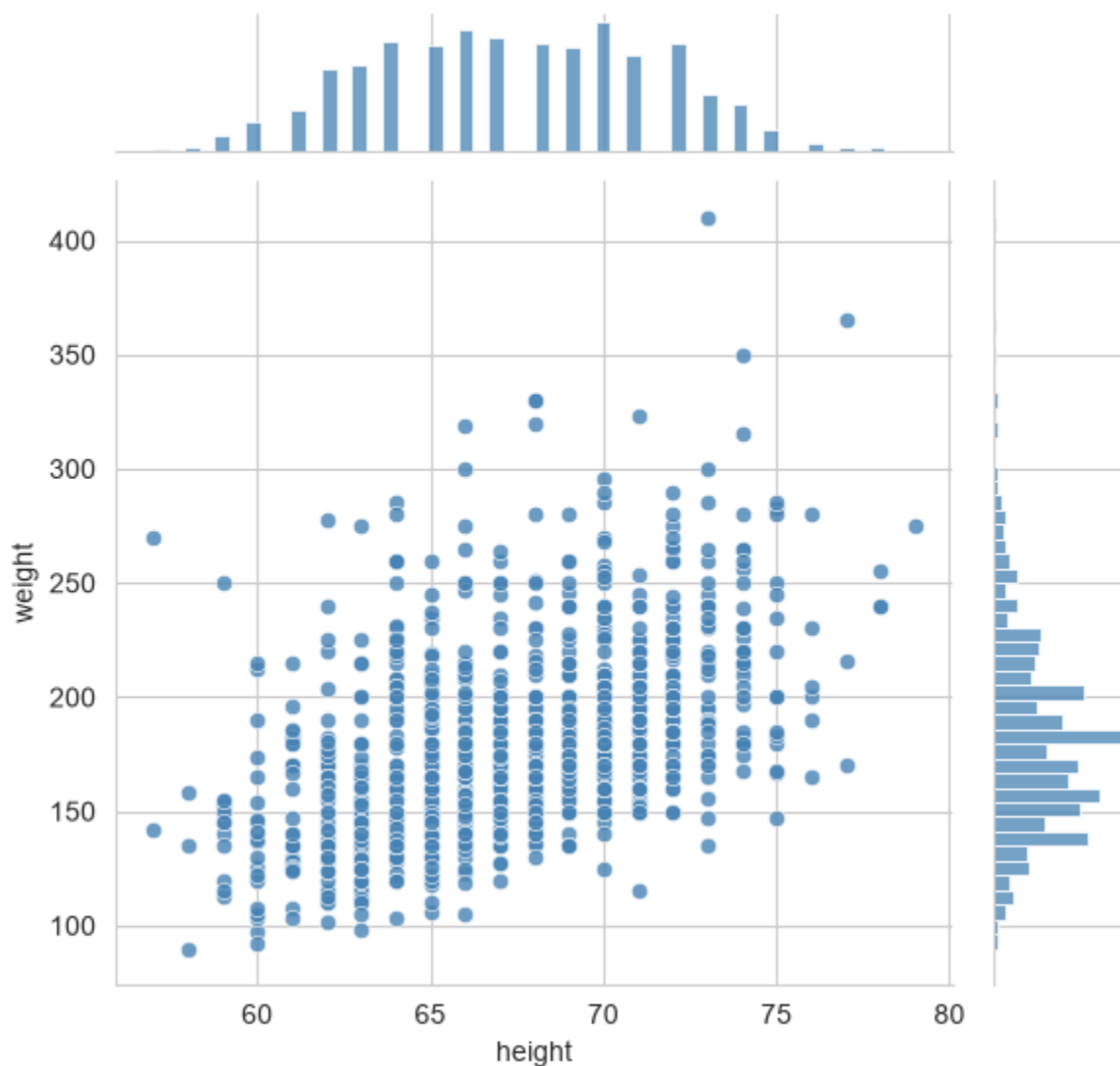


2.5. Joint Distribution PLOT of Weight Against Height

In [13]:

```
joint_plot = sns.jointplot(data = df, x = "height", y = "weight",
                           kind = 'scatter', alpha = 0.8, color = 'steelblue',
                           marginal_kws = dict(bins = 50, fill = True)
                           )
joint_plot.fig.suptitle('A Joint Plot of Weight and Height Distribution',
                        fontsize = 14, fontweight = 'bold')
plt.tight_layout()
plt.show()
```


A Joint Plot of Weight and Height Distribution



In [14]:

```
print('='*110)
print(f"Correlation Between Weight and Height")
print('='*110)
corr_val = df['height'].corr(df['weight'])
print(f"Correlation: {corr_val:.3f}")
```

```
=====
Correlation Between Weight and Height
=====
```

```
Correlation: 0.482
```

3. Categorical Variables Analysis

3.1. Bar Showing the Distribution of Gender

In [15]:

```
df['sex'].value_counts()
```

Out[15]:

```
sex
MALE      507
FEMALE    487
Name: count, dtype: int64
```

In [16]:

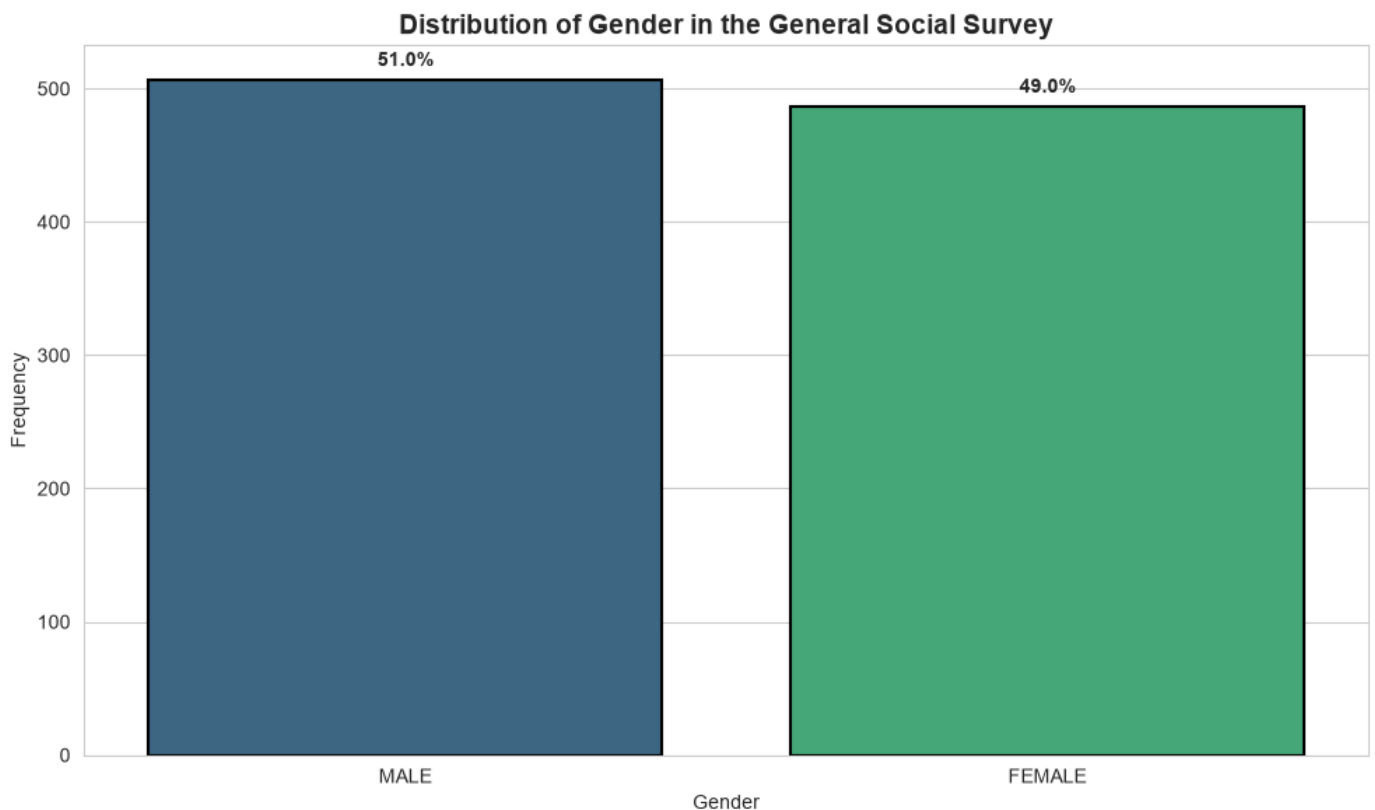
```
plt.figure(figsize = (10,6))
ax = sns.countplot(data = df, x = "sex", palette = 'viridis',
                  edgecolor = 'black', linewidth = 1.5)

plt.title("Distribution of Gender in the General Social Survey",
         fontsize = 14,
         fontweight = 'bold')

plt.xlabel("Gender")
plt.ylabel("Frequency")

total = len(df)
for p in ax.patches:
    percentage = f'{100*p.get_height()/total:.1f}%'
    x = p.get_x() + p.get_width()/2
    y = p.get_height()+10
    ax.annotate (percentage, (x, y),
                ha = 'center',
                fontweight = 'bold')

plt.tight_layout()
plt.show()
```



3.2. Bar Showing the Distribution of Degree Qualification

In [17]:

```
Frequency_table = df['degree'].value_counts().sort_values()
Frequency_table
```

Out[17]:

```
degree
LT HIGH SCHOOL      77
JUNIOR COLLEGE      92
GRADUATE            125
BACHELOR            209
HIGH SCHOOL         491
Name: count, dtype: int64
```

In [18]:

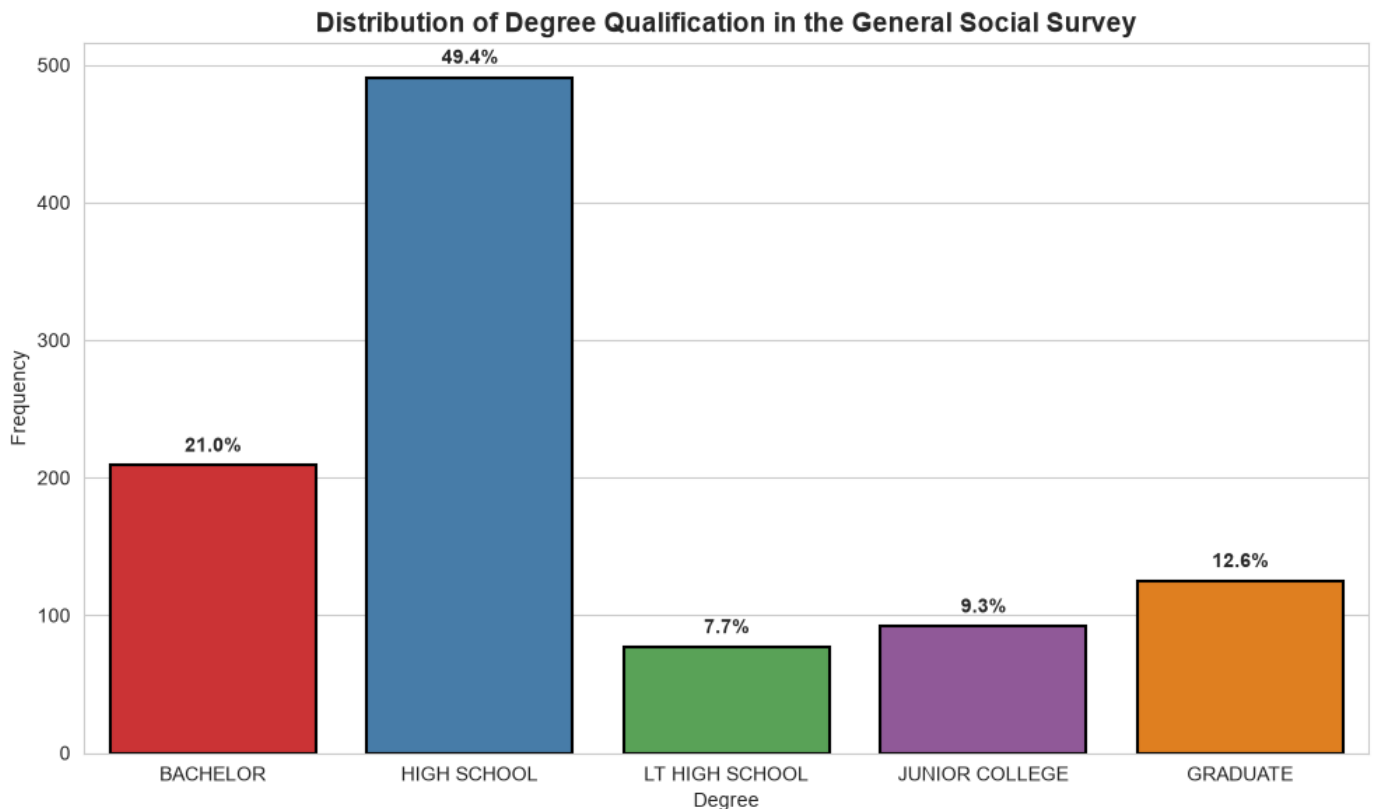
```
plt.figure(figsize = (10,6))
ax = sns.countplot(data = df, x = "degree", palette = 'Set1',
                  edgecolor = 'black', linewidth = 1.5)

plt.title("Distribution of Degree Qualification in the General Social Survey",
          fontsize = 14,
          fontweight = 'bold')

plt.xlabel("Degree")
plt.ylabel("Frequency")

total = len(df)
for p in ax.patches:
    percentage = f'{100*p.get_height()/total:.1f}%'
    x = p.get_x() + p.get_width()/2
    y = p.get_height()+10
    ax.annotate (percentage, (x, y),
                ha = 'center',
                fontweight = 'bold')

plt.tight_layout()
plt.show()
```



3.2. Bar Showing the Distribution of Marital Status

In [19]:

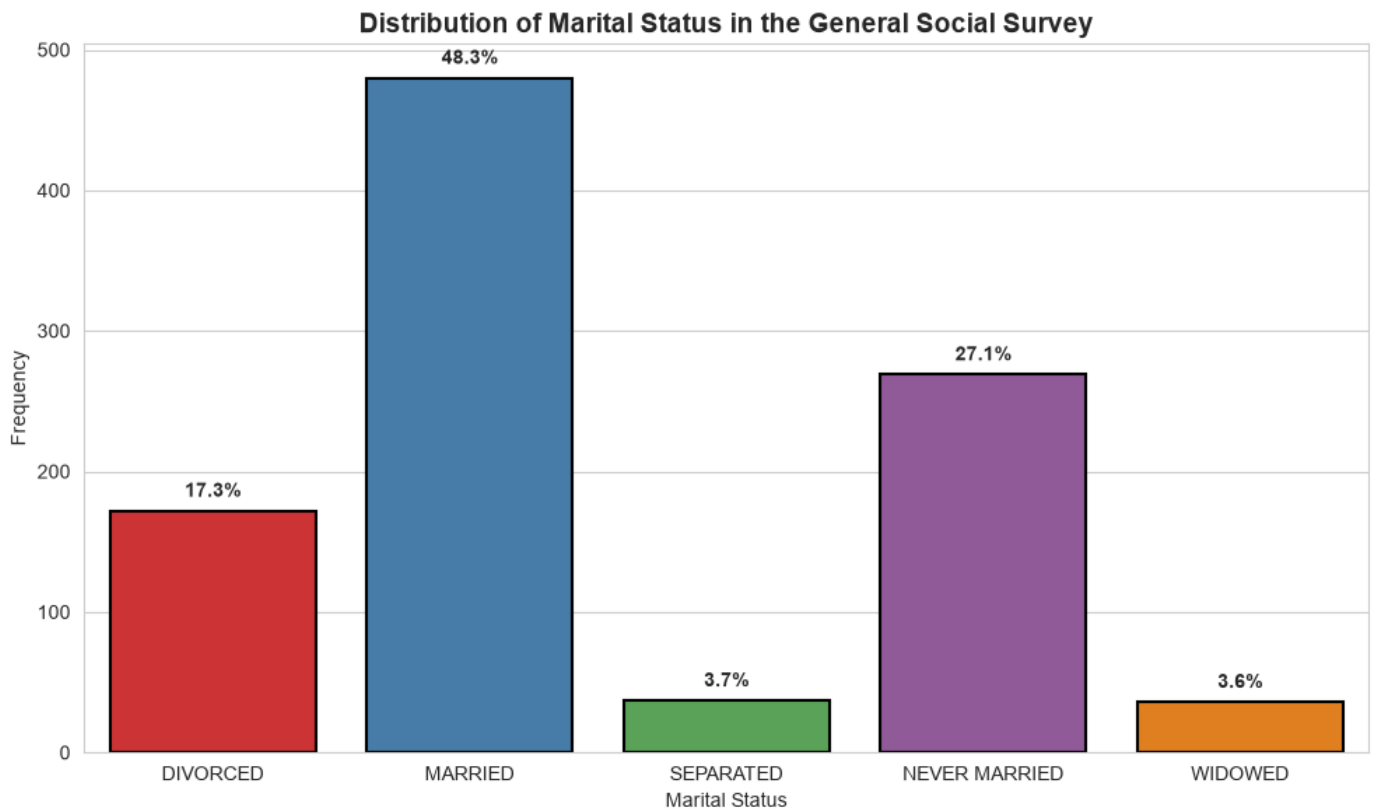
```
plt.figure(figsize = (10,6))
ax = sns.countplot(data = df, x = "marital", palette = 'Set1',
                  edgecolor = 'black', linewidth = 1.5)

plt.title("Distribution of Marital Status in the General Social Survey",
          fontsize = 14,
          fontweight = 'bold')

plt.xlabel("Marital Status")
plt.ylabel("Frequency")

total = len(df)
for p in ax.patches:
    percentage = f'{100*p.get_height()/total:.1f}%'
    x = p.get_x() + p.get_width()/2
    y = p.get_height()+10
    ax.annotate (percentage, (x, y),
                ha = 'center',
                fontweight = 'bold')

plt.tight_layout()
plt.show()
```



3.3. Horizontal Bar Chart Showing the Distribution of Degree Qualification

In [36]:

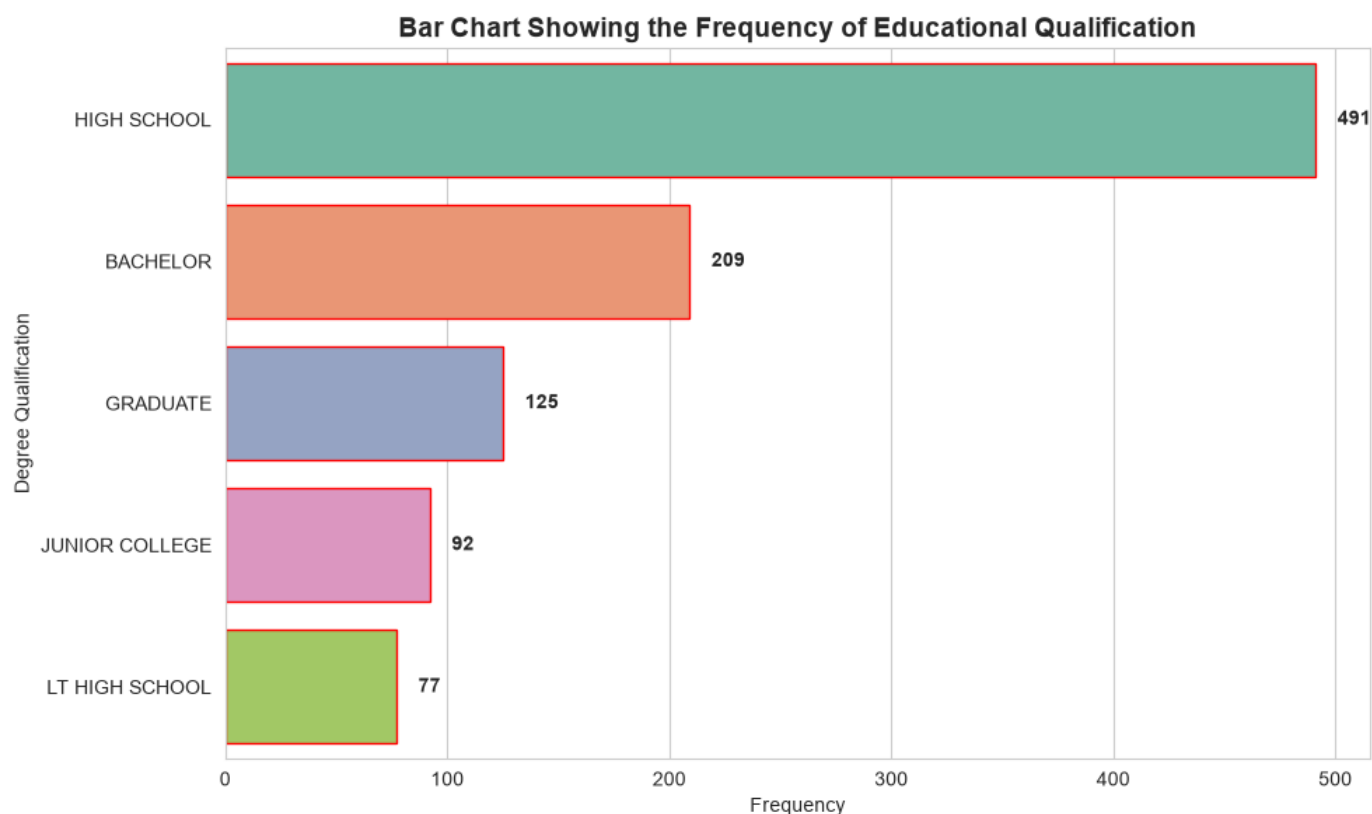
```
plt.figure(figsize = (10,6))
degree_counts = df['degree'].value_counts()
ax = sns.barplot(x = degree_counts.values, y = degree_counts.index,
                 palette = 'Set2', edgecolor = 'red')
plt.title("Bar Chart Showing the Frequency of Educational Qualification",
          fontsize = 14, fontweight = 'bold')
plt.xlabel("Frequency")
plt.ylabel("Degree Qualification")

for i, v in enumerate(degree_counts.values):
    ax.text(v+10, i, str(v), va = 'center',
            fontweight = 'bold')

plt.tight_layout()
plt.show

print(degree_counts)
```

```
degree
HIGH SCHOOL    491
BACHELOR       209
GRADUATE       125
JUNIOR COLLEGE  92
LT HIGH SCHOOL  77
Name: count, dtype: int64
```



3.4. Pie Chart Showing the Distribution of Degree Qualification

In [53]:

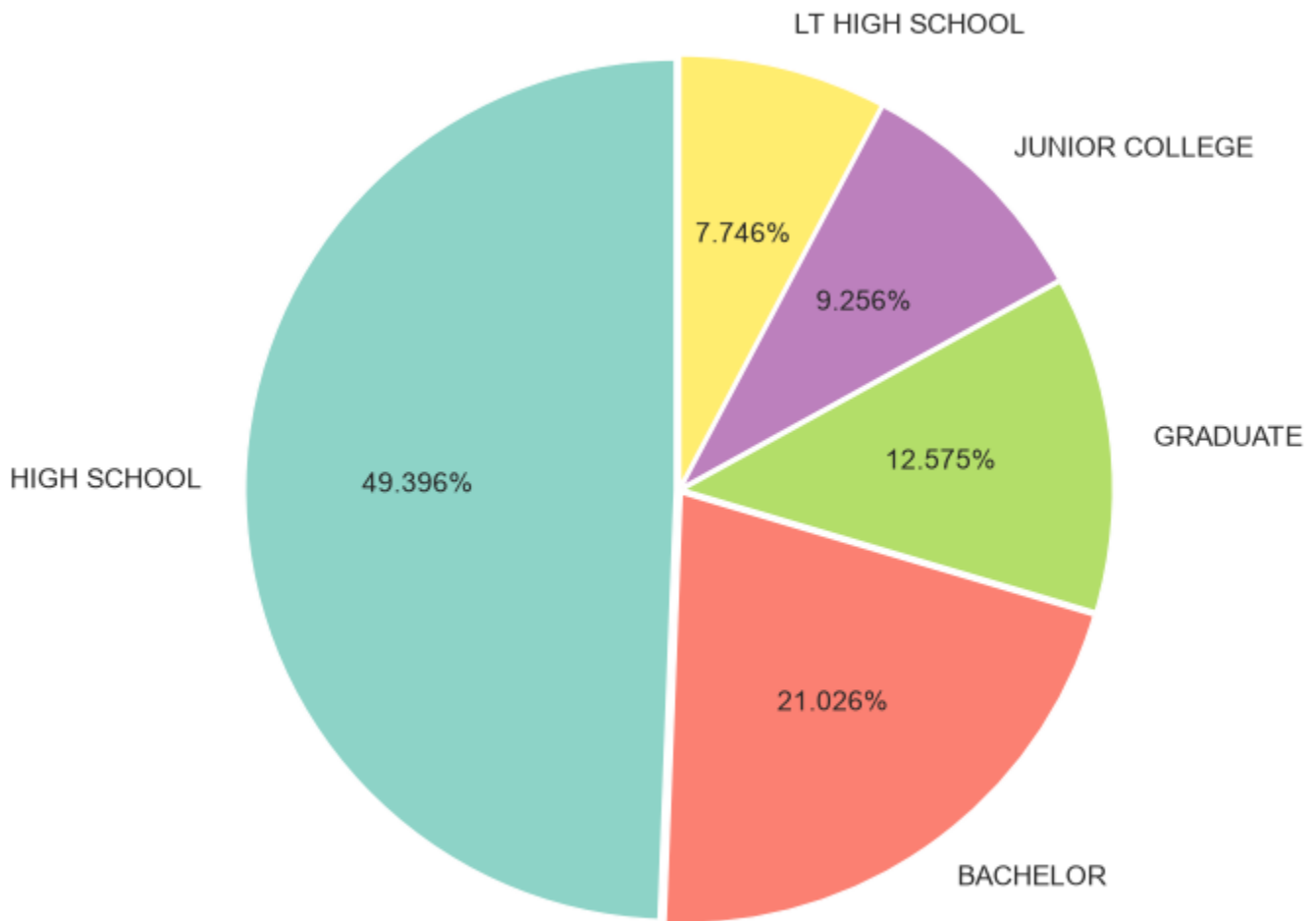
```
plt.figure(figsize = (14, 6))
degree_counts = df['degree'].value_counts()

color = plt.cm.Set3(np.linspace(0,1,len(degree_counts)))
#color

plt.pie(
    degree_counts,
    labels = degree_counts.index,
    autopct = '%1.3f%%',
    colors = color,
    startangle = 90,
    explode = [0.01]*len(degree_counts)
)

plt.title("Pie Chart Showing the Distribution of Degree Qualification",
          fontsize = 14,
          fontweight = 'bold')
plt.tight_layout()
plt.show()
```

Pie Chart Showing the Distribution of Degree Qualification



3.5. Pie Chart and Bar Chart for Degree Qualification

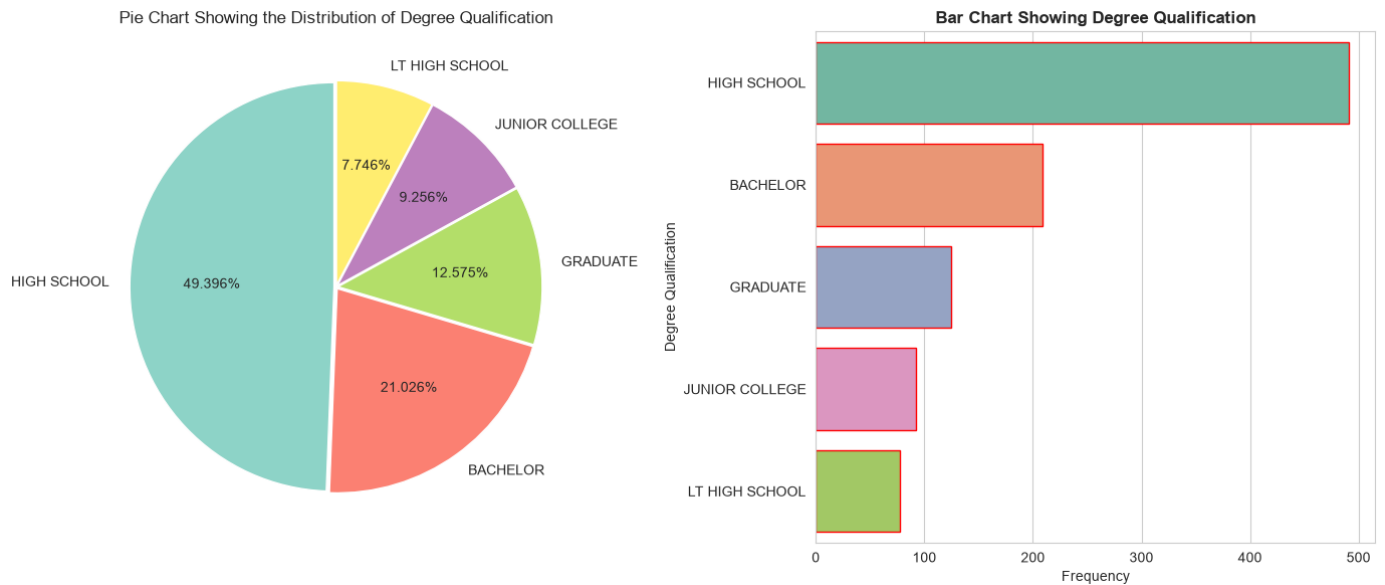
In [60]:

```
fig, axes = plt.subplots(1,2, figsize = (14, 6))
degree_counts = df['degree'].value_counts()
color = plt.cm.Set3(np.linspace(0,1,len(degree_counts)))

axes[0].pie(degree_counts, labels = degree_counts.index,
            autopct = '%1.3f%%',
            colors = color,
            startangle = 90,
            explode = [0.01]*len(degree_counts))
axes[0].set_title("Pie Chart Showing the Distribution of Degree Qualification")

sns.barplot(x = degree_counts.values, y = degree_counts.index, ax = axes[1],
            palette = 'Set2', edgecolor = 'red')
axes[1].set_title("Bar Chart Showing Degree Qualification",
                  fontweight = 'bold',
                  fontsize = 12)
axes[1].set_xlabel("Frequency")
axes[1].set_ylabel("Degree Qualification")
```

```
plt.tight_layout()
plt.show()
```



2 by 2 Plots- Four Chart is one Plot

In [83]:

```
# Style
sns.set_style("whitegrid")

# Degree frequencies
degree_counts = df['degree'].value_counts()
colors = plt.cm.Set3(np.linspace(0, 1, len(degree_counts)))

# Create 2x2 layout
fig, axes = plt.subplots(2, 2, figsize=(15, 12))

# -----
# 1. Pie Chart
# -----
axes[0, 0].pie(
    degree_counts,
    labels=degree_counts.index,
    autopct='%1.1f%%',
    startangle=90,
    colors=colors,
    explode=[0.03] * len(degree_counts)
)
axes[0, 0].set_title(
    "Distribution of Degree Qualification",
    fontsize=13,
    fontweight='bold'
)

# -----
# 2. Horizontal Bar Chart
# -----
sns.barplot(
    x=degree_counts.values,
    y=degree_counts.index,
    palette='Set2',
```



```

    edgecolor='black',
    ax=axes[0, 1]
)

axes[0, 1].set_title(
    "Frequency of Degree Qualification",
    fontsize=13,
    fontweight='bold'
)
axes[0, 1].set_xlabel("Frequency")
axes[0, 1].set_ylabel("Degree")

# Add frequency labels
for i, v in enumerate(degree_counts.values):
    axes[0, 1].text(v + 2, i, str(v), va='center', fontsize=10)

# -----
# 3. Histogram of Income
# -----
sns.histplot(
    data=df,
    x='income',
    bins=20,
    kde=True,
    color='steelblue',
    edgecolor='black',
    ax=axes[1, 0]
)

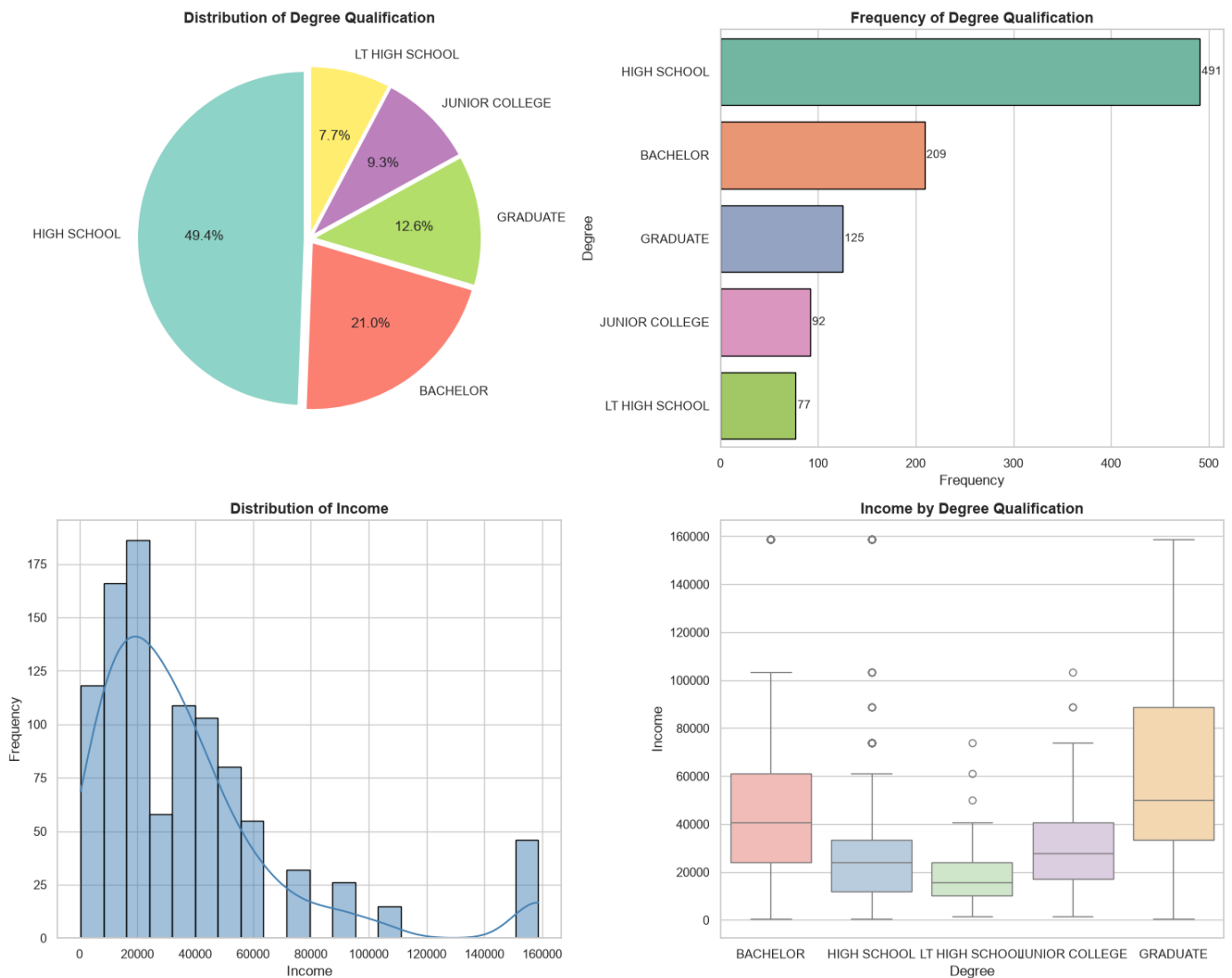
axes[1, 0].set_title(
    "Distribution of Income",
    fontsize=13,
    fontweight='bold'
)
axes[1, 0].set_xlabel("Income")
axes[1, 0].set_ylabel("Frequency")

# -----
# 4. Boxplot: Income by Degree
# -----
sns.boxplot(
    data=df,
    x='degree',
    y='income',
    palette='Pastel1',
    ax=axes[1, 1]
)

axes[1, 1].set_title(
    "Income by Degree Qualification",
    fontsize=13,
    fontweight='bold'
)
axes[1, 1].set_xlabel("Degree")
axes[1, 1].set_ylabel("Income")

plt.tight_layout()
plt.show()

```



Violin PLots

Violin Plot: Income by Degree

In [88]:

```
import matplotlib.pyplot as plt
import seaborn as sns

# Set publication-quality theme
sns.set_theme(style="whitegrid", context="talk")

plt.figure(figsize=(10, 6))

sns.violinplot(
    data=df,
    x="degree",
    y="income",
    palette="viridis",
    inner="box",      # Shows median and quartiles
    linewidth=1.2
)

plt.title("Distribution of Income by Educational Degree",
```

```

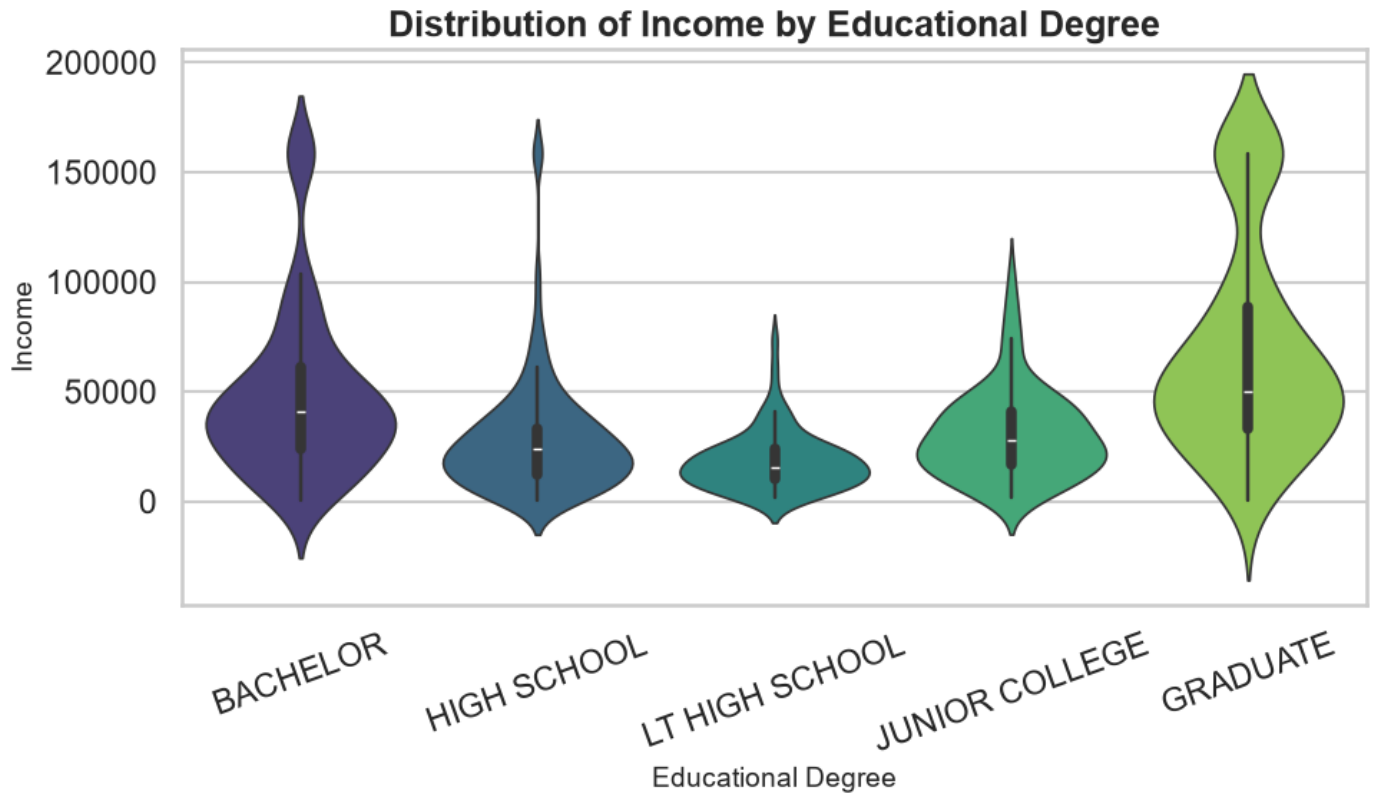
        fontsize=18, fontweight="bold")

plt.xlabel("Educational Degree", fontsize=14)
plt.ylabel("Income", fontsize=14)

plt.xticks(rotation=20)

plt.tight_layout()
plt.show()

```



Violin Plot: Income by Degree, Split by Sex

In [89]:

```

plt.figure(figsize=(11, 6))

sns.violinplot(
    data=df,
    x="degree",
    y="income",
    hue="sex",
    split=True,          # Works only if sex has two categories
    inner="quartile",
    palette="Set2",
    linewidth=1.2
)

plt.title("Distribution of Income by Degree and Sex",
          fontsize=18, fontweight="bold")

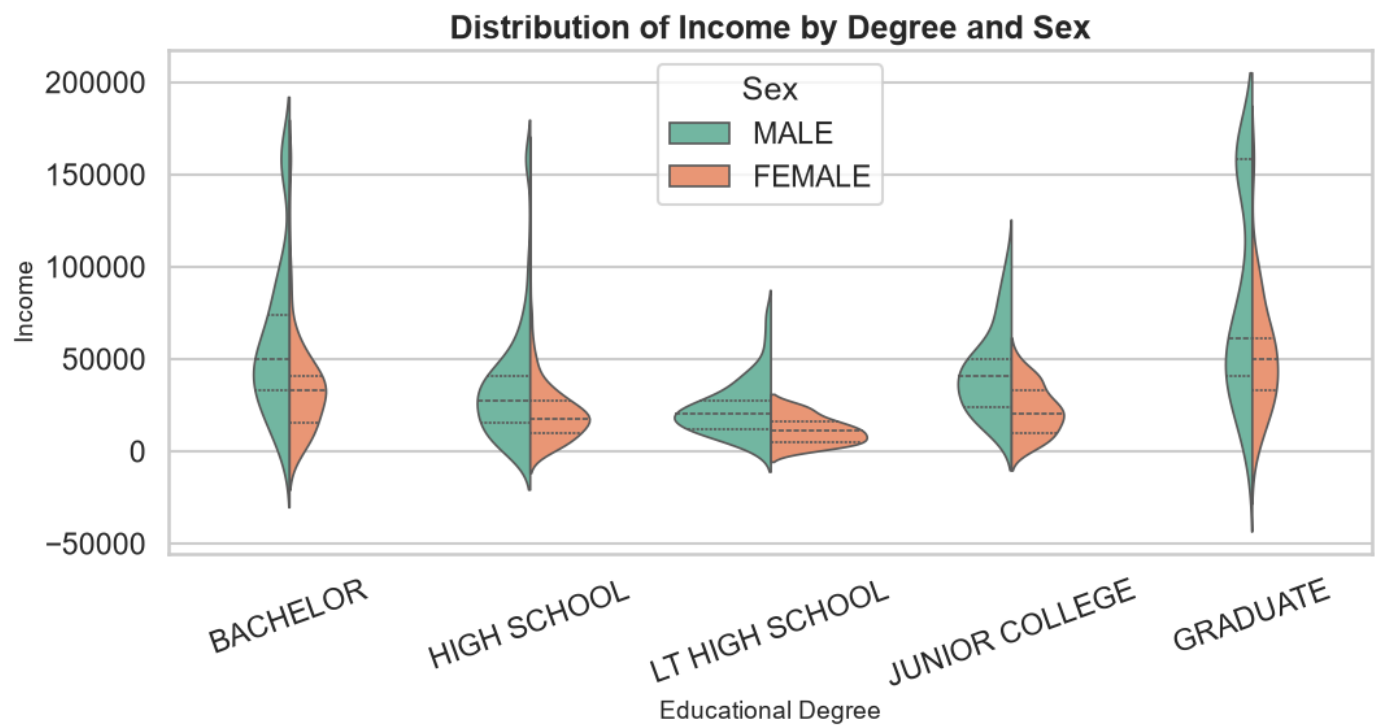
plt.xlabel("Educational Degree", fontsize=14)
plt.ylabel("Income", fontsize=14)

plt.legend(title="Sex")

```

```
plt.xticks(rotation=20)
```

```
plt.tight_layout()  
plt.show()
```



In [94]:

```
plt.figure(figsize=(11, 6))
```

```
sns.violinplot(  
    data=df,  
    x="degree",  
    y="income",  
    hue="sex",  
    dodge=True,  
    inner=None,  
    palette="Set2"  
)
```

```
sns.stripplot(  
    data=df,  
    x="degree",  
    y="income",  
    hue="sex",  
    dodge=True,  
    color="black",  
    alpha=0.35,  
    size=1  
)
```

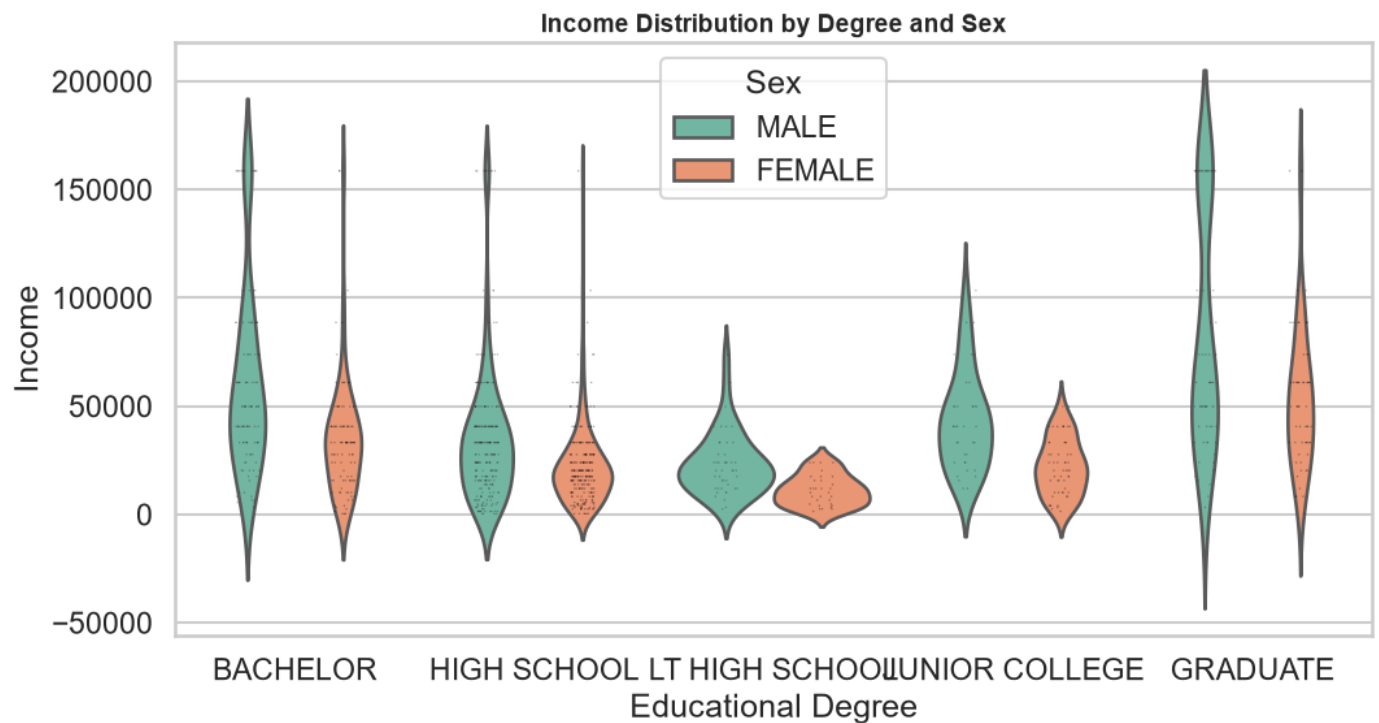
```
# Remove duplicate legend
```

```
handles, labels = plt.gca().get_legend_handles_labels()  
plt.legend(handles[:2], labels[:2], title="Sex")
```

```
plt.title("Income Distribution by Degree and Sex",  
         fontsize=14,  
         fontweight="bold")
```

```
plt.xlabel("Educational Degree")
plt.ylabel("Income")

plt.tight_layout()
plt.show()
```



Correlational Heatmap

In [85]:

```
# Select continuous variables
continuous_vars = ['age', 'income', 'hrswrk', 'height', 'weight']

# Compute correlation matrix
corr_matrix = df[continuous_vars].corr(method='pearson')

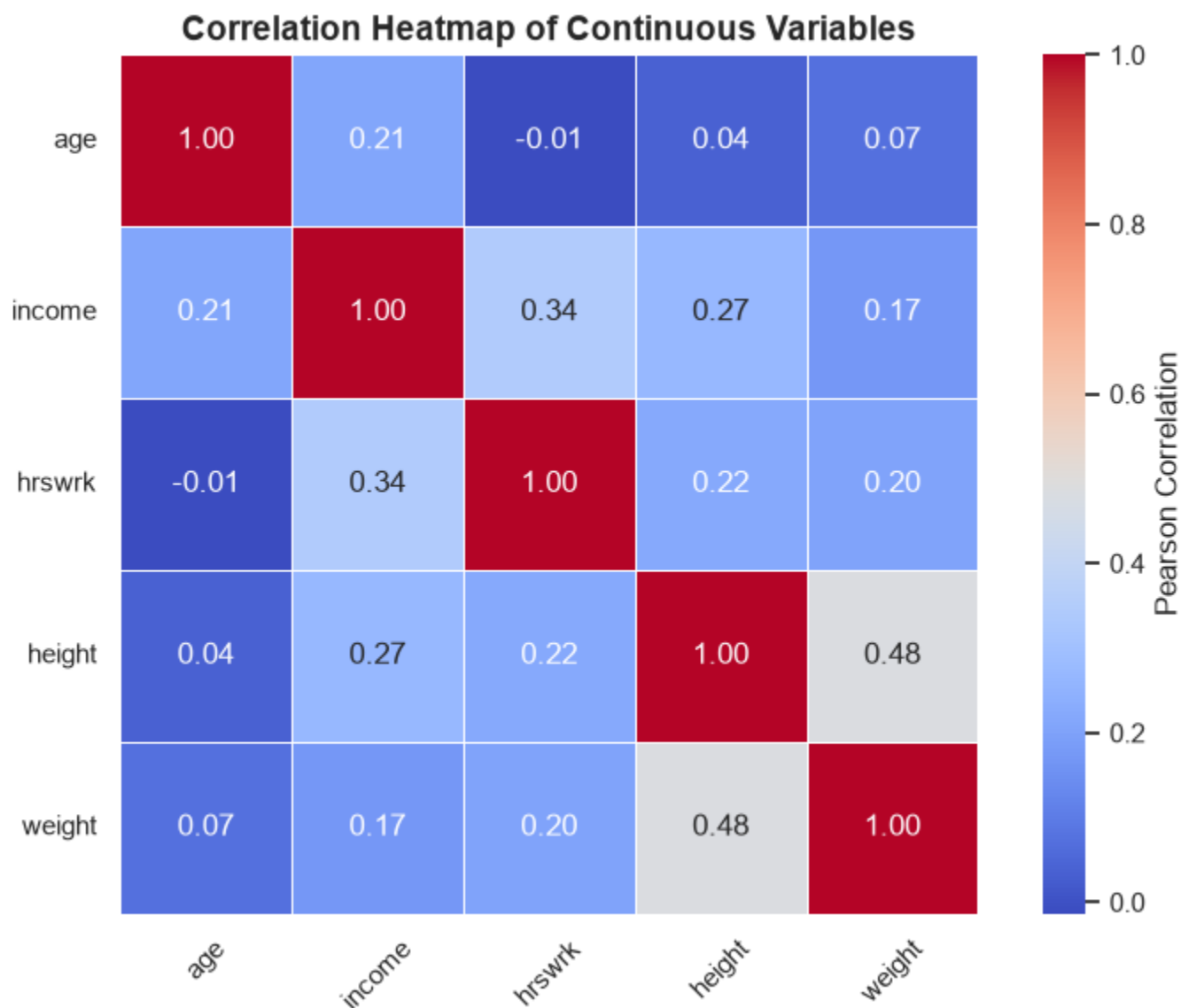
# Plot heatmap
plt.figure(figsize=(8, 6))

sns.heatmap(
    corr_matrix,
    annot=True,           # Display correlation coefficients
    cmap='coolwarm',     # Color palette
    fmt='.2f',           # Two decimal places
    linewidths=0.5,
    square=True,
    cbar_kws={'label': 'Pearson Correlation'}
)

plt.title(
    'Correlation Heatmap of Continuous Variables',
    fontsize=14,
    fontweight='bold'
)

plt.xticks(rotation=45)
plt.yticks(rotation=0)
```

```
plt.tight_layout()
plt.show()
```



Pairplot

In [86]:

```
# Continuous variables
continuous_vars = ['age', 'income', 'hrswrk', 'height', 'weight']

# Pairplot
pair = sns.pairplot(
    df[continuous_vars],
    diag_kind='hist',      # Histogram on the diagonal
    corner=True,          # Plot only the lower triangle
    plot_kws={
        'alpha': 0.7,
        's': 40,
        'edgecolor': 'black'
    },
    diag_kws={
        'bins': 20,
        'edgecolor': 'black'
    }
)
```

```

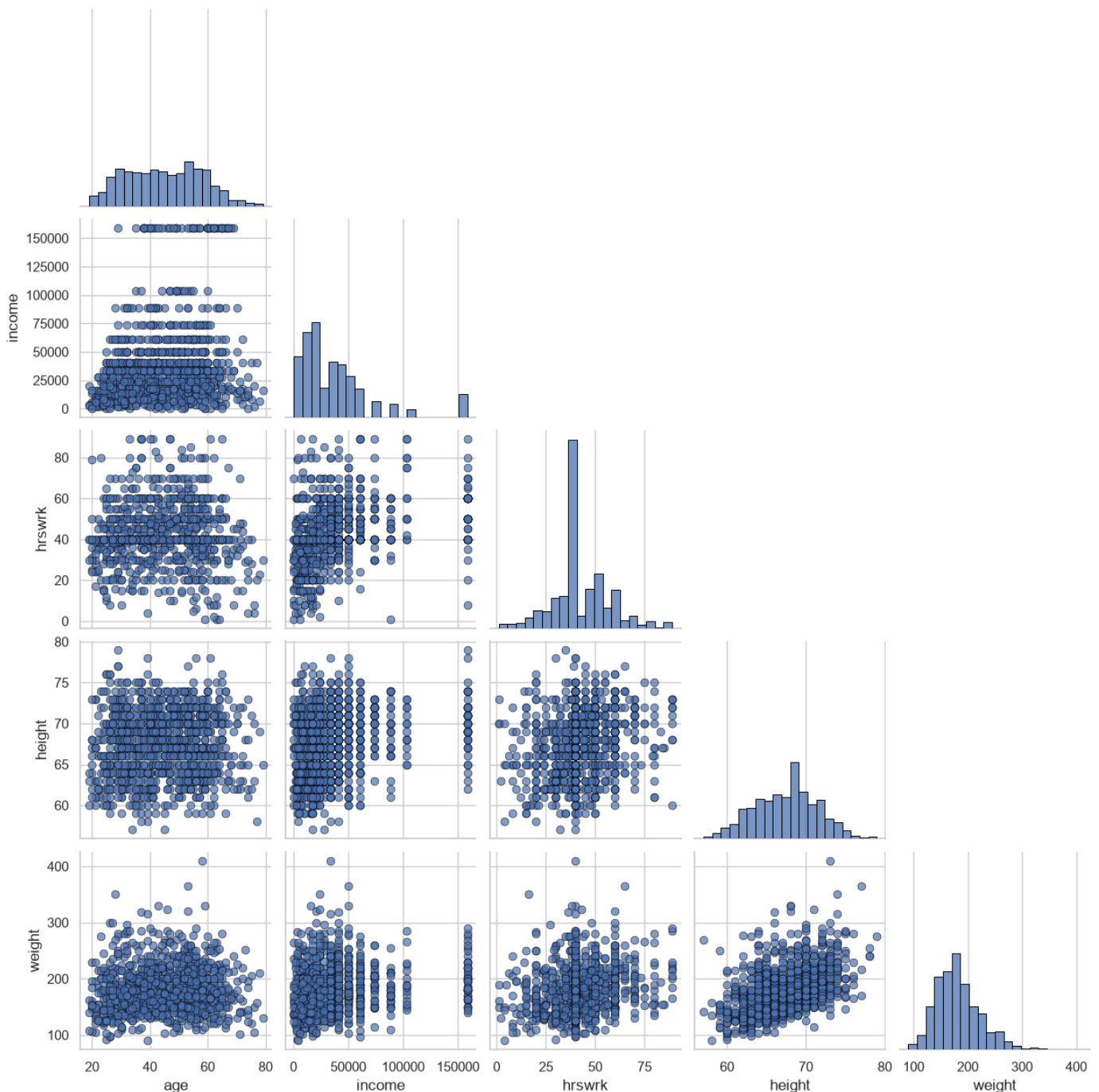
    }
)

pair.fig.suptitle(
    'Pairwise Relationships Among Continuous Variables',
    fontsize=16,
    fontweight='bold',
    y=1.02
)

plt.show()

```

Pairwise Relationships Among Continuous Variables



3.6. Income by Degree Qualification

In [82]:

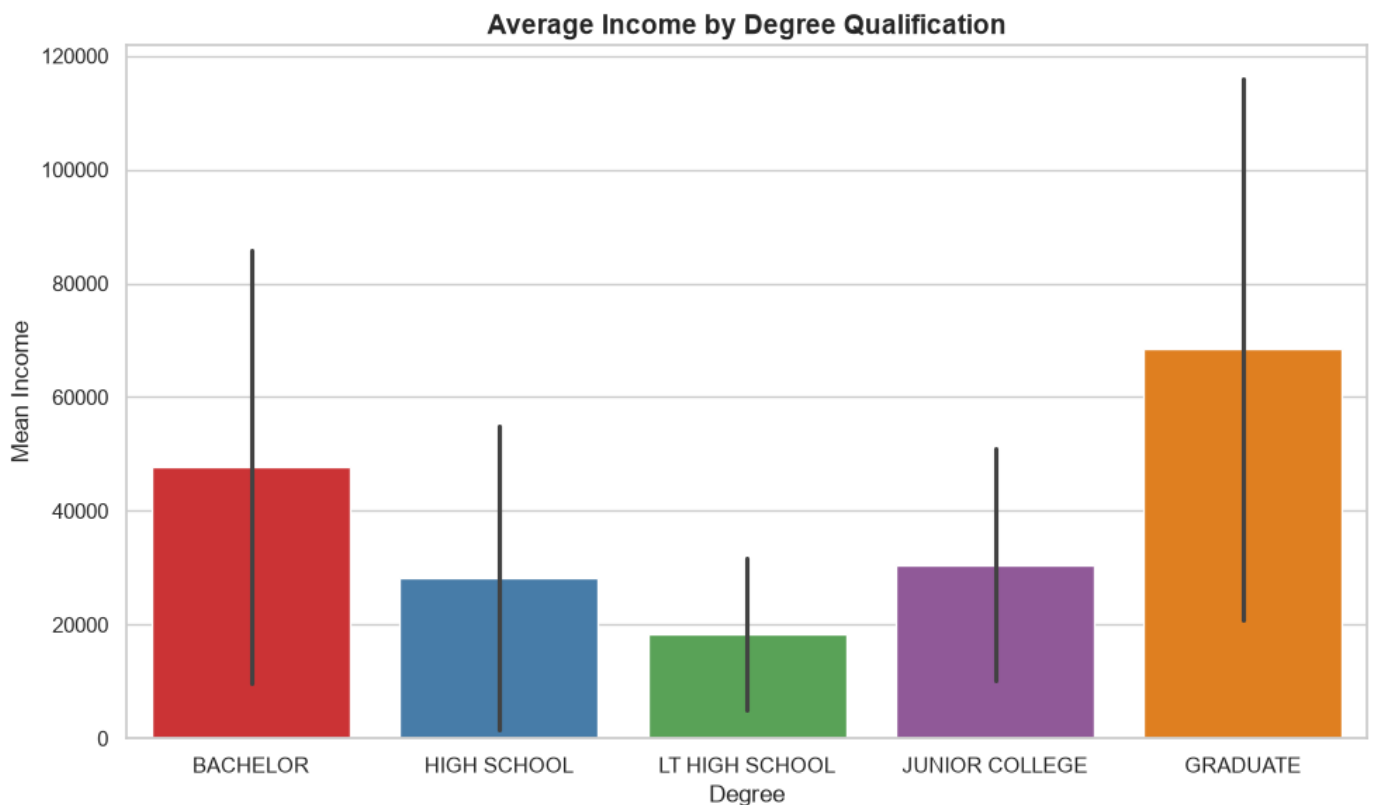
```

sns.set_theme(style = 'whitegrid')
plt.figure(figsize = (10, 6))

sns.barplot(data = df,
            x = 'degree',
            y = 'income',
            estimator = 'mean',
            errorbar='sd',
            palette = 'Set1')
plt.title("Average Income by Degree Qualification",
          fontsize = 14,
          fontweight = 'bold')
plt.xlabel("Degree")
plt.ylabel("Mean Income")
plt.xticks(rotation = 0)
plt.tight_layout()

plt.show()

```



3.7. Comparative Bar Chart of Income by Degree Qualification Across Gender

In [75]:

```

sns.set_theme(style = 'whitegrid')
plt.figure(figsize = (10, 6))

sns.barplot(data = df,
            x = 'degree',
            y = 'income',
            hue = 'sex',
            estimator = 'mean',
            errorbar = 'sd',
            palette = 'Set1')

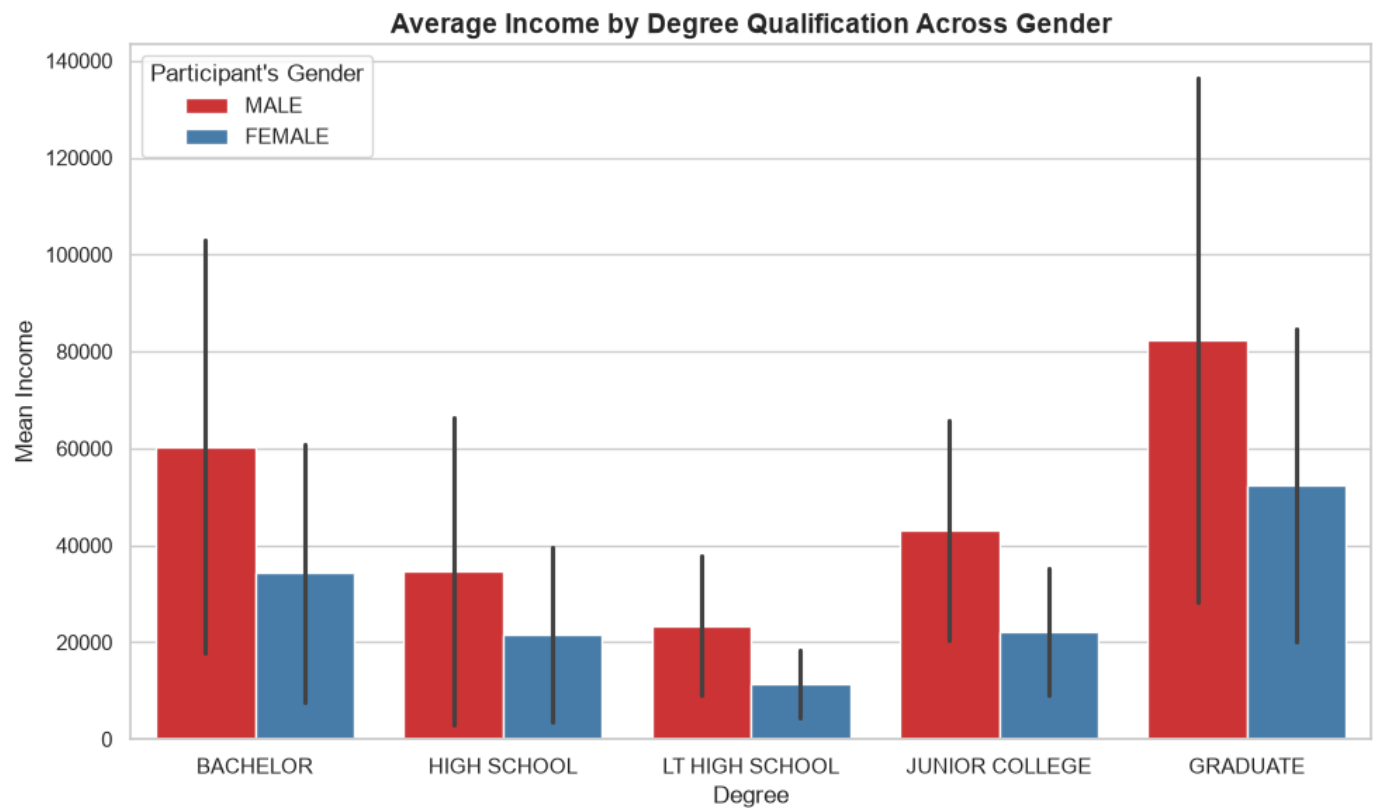
```



```
plt.title("Average Income by Degree Qualification Across Gender",
          fontsize = 14,
          fontweight = 'bold')
plt.xlabel("Degree")
plt.ylabel("Mean Income")

plt.legend(title = "Participant's Gender")
plt.xticks(rotation = 0)
plt.tight_layout()

plt.show()
```



In [76]:

```
plt.figure(figsize=(11,7))

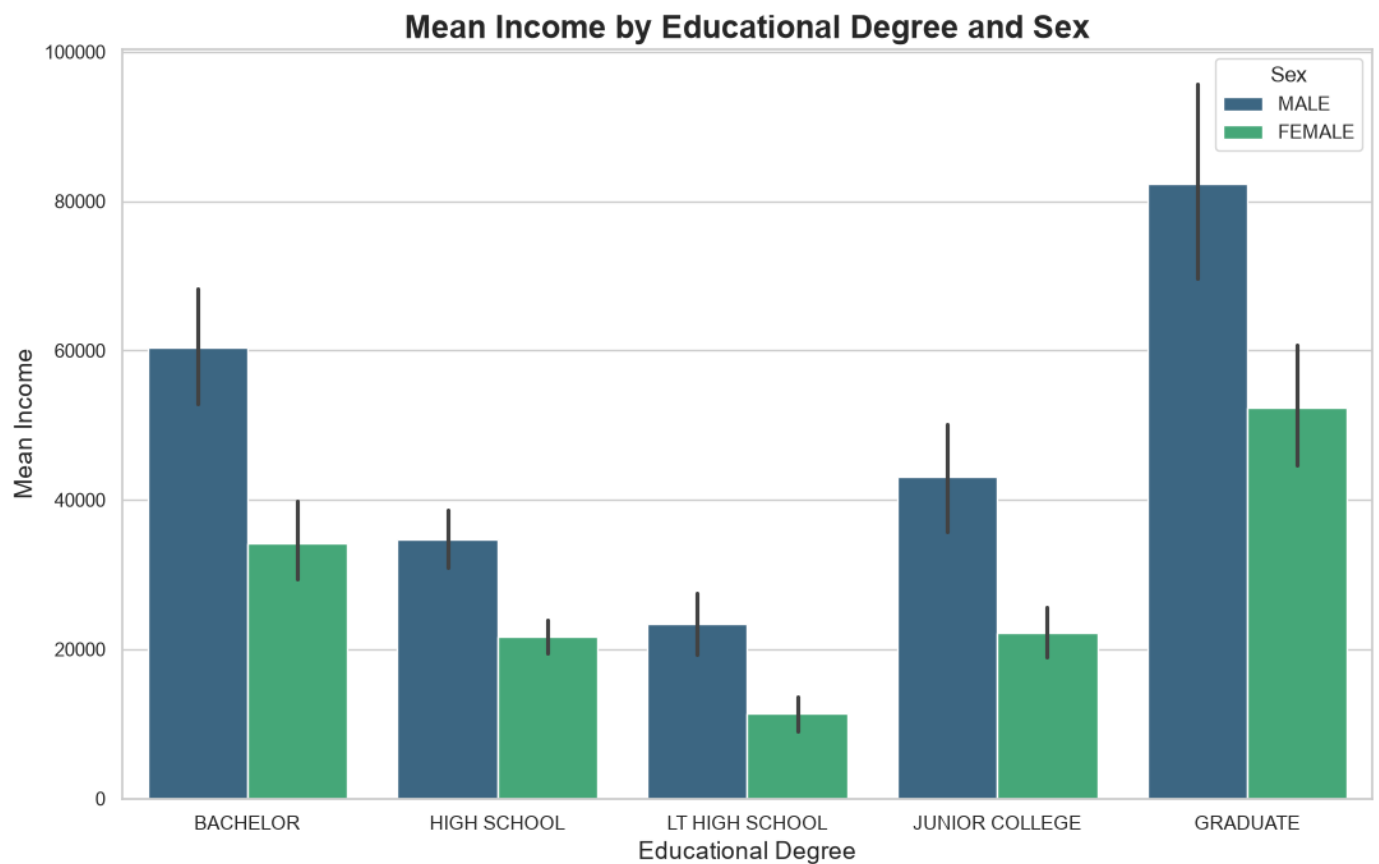
ax = sns.barplot(
    data=df,
    x="degree",
    y="income",
    hue="sex",
    estimator="mean",
    errorbar=("ci", 95),
    palette="viridis"
)
# Improve appearance
ax.set_title(
    "Mean Income by Educational Degree and Sex",
    fontsize=18,
    fontweight="bold"
)

ax.set_xlabel("Educational Degree", fontsize=14)
ax.set_ylabel("Mean Income", fontsize=14)
```

```
plt.xticks(fontsize=11)
plt.yticks(fontsize=11)

plt.legend(
    title="Sex",
    title_fontsize=12,
    fontsize=11
)

sns.despine
plt.tight_layout()
plt.show()
```



THE END

DATA VISUALIZATION AND INSIGHTS EXPLAINED

*From Data to Knowledge,
from Knowledge to Decisions*

Thank\ You!



In []: